Seventeen Years in the Making: Tracing Long-Term Design in Assistive Robotic Furnishings

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Abstract

Human-centered robotics is often reported through individual prototypes, studies, and funded projects. While longitudinal HRI examines how interaction with robots develops over time, a complementary question concerns how a design inquiry persists as its artifacts and research conditions change. We examine this question through the 17-year trajectory of home+, a research program investigating robotic furnishings and environments for aging in place. Drawing on publications, a master's thesis, technical and design records, prototype documentation, and written retrospective accounts from two long-term contributors, we reconstruct selected instantiations from 2009 to 2026. Rather than interpreting the trajectory solely as successive versions of one robot, we examine relationships among changing realizations of a continuing inquiry. The analysis identifies continuity that does not depend on preserving a single artifact, including branching and re-convergence, shifts in the unit and fidelity of design, technical specialization, and repeated re-embodiment. The retrospective accounts further highlight how researcher handoffs combine reconstruction with redirection, how opportunities enable continuation while shaping its direction, and how institutional moves require rebuilding local capacity for HRI. Together, these findings characterize long-term design as the maintenance and revision of an inquiry across changing material and organizational conditions. We discuss implications for documenting transitions, supporting inheritance across researcher generations, and distinguishing persistence from evidence of assistive success.

CCS Concepts

• **Human-centered computing → Empirical studies in HCI**.

Keywords

human-robot interaction, assistive robotics, design research, long-term design, robotic furnishings, aging in place

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Figure 1: The 2026 home+ prototype in a posed illustration of household assistance. The photo depicts an intended interaction, not a user-study session.

1 Introduction

Some stories can only be told across years. In a design history, the relationships among artifacts can reveal changes that no individual prototype report captures on its own: a concern can persist while its physical realization changes, a technical problem can become a research strand, and an apparently settled question can be reopened through a different form of prototyping. The value of an extended account lies not in duration, but in what comparison across that duration makes visible. In home+, this includes retuning control after a lamp rebuild and returning to full-scale mock-ups to examine how people might command and encounter robotic assistance [2, 11].

Longitudinal HRI examines interaction beyond initial encounters, including sustained or repeated use of robots [16, 18]. A complementary question concerns the timescale of designing those robots. Retrospective accounts of *Interaction Composer* and assistive robotics programs already demonstrate the value of learning across years of practice [3, 7]. Programmatic design research likewise examines how particular designs both express and reshape a broader inquiry [14]. Building on these precedents, we ask *what maintaining an assistive robotics inquiry involves* when its embodiments, representations, researchers, and research conditions change.

We examine this question through the seventeen-year trajectory of home+, a research program investigating robotic furnishings and

environments for aging in place. The comforTABLE concept introduced at HRI 2009 proposed a networked assistive environment that would support people's own activity by reorganizing their immediate environment [9]. Later work pursued related concerns through the development of a table for the clinical environment, distributed and cooperating furnishings, technical investigations of a continuum lamp, and full-scale nonfunctional representations [2, 4, 12, 23, 24]. The current realization combines camera-based gesture interaction with mobile manipulation for household assistance. Figure 1 illustrates the intended household context of the current realization.

We use *long-term design* to describe the persistence and revision of an inquiry across extended periods and multiple realizations. Rather than interpreting home+ solely as successive versions of one robot, we examine relationships among actual build generations, parallel research strands, and changing ways of investigating a recurring concern: how robotic furnishings and environments might support everyday activity while preserving independence and agency. Our guiding question is: *What becomes visible about human-centered robot design when an inquiry persists across multiple generations of artifacts and research activity?*

We reconstruct selected realizations from 2009 to 2026 using publications, a master's thesis, technical and design records, prototype documentation, and written retrospective accounts from two long-term contributors. The paper makes two contributions. First, it provides a sourced reconstruction connecting the selected realizations, the questions they addressed, and their reported limitations. Second, it develops six lessons about changing the unit of design, re-embodiment, prototype fidelity, researcher handoffs, opportunities for continuation, and institutional moves. Together, these contributions explain how continuity was maintained and revised, including reconstruction costs and unresolved questions that are less visible within a single prototype or project cycle.

2 Related Work

Robotic furnishings connect physical function to the design of interaction. Sirkin et al.'s Mechanical Ottoman examines how a moving footstool offers and withdraws support [21]. Roombots explores modular robots intended to form furniture that changes shape and location [22]. These projects provide related examples of adaptive furnishings; our focus is how an inquiry into such assistance changes across a longer history of designs and research.

2.1 Long-Term HRI and Retrospective Design

Long-term HRI examines how interaction develops beyond initial encounters. Leite et al. survey sustained interaction with social robots, while Matheus et al. distinguish repeated interaction with the same users from extended deployments involving changing users [16, 18]. Our temporal object is instead the *design inquiry*: how related questions are pursued through changing robotic artifacts and research activities. Deployment can inform redesign, but seventeen years of design activity does not imply seventeen years of use.

Retrospective design accounts also have precedents within HRI. Glas et al. examine eight years of *Interaction Composer*, a visual flow-based tool for programming social robot interactions, through interaction flows, case studies, and user interviews [7]. They derive lessons about reuse, event handling, and collaboration between programmers and non-programmers, alongside limitations of the authoring framework. Their contribution therefore extends beyond describing their tool. Its primary focus, however, is the framework and the design practices it supports. The home+ history examines a different unit of continuity: an assistive-robotics inquiry pursued through successive and parallel physical realizations, changing research emphases, and organizational transitions.

Cortellessa et al. retrospectively examine almost eighteen years of AI and robotics projects for older adults, connecting project experiences to lessons for deployment and acceptance through technological, socio-health, and human perspectives [3]. Their account provides a domain-specific precedent for learning across research projects.

2.2 Design Programs and Emergent Practice

Programmatic design research connects individual design engagements to a broader inquiry. Löwgren et al. describe an interplay between the perspectives brought to a program and the engagements through which knowledge develops [17]. Particular artifacts can therefore express existing commitments and prompt reconsideration. Redström similarly describes design programs as bringing together definitions developed through design practice [20].

Höök et al.'s retrospective on more than a decade of soma design examines changes in learning, representations, scope, and underlying commitments [14]. It considers generativity, transferability, and scope in relation to the program's own worldview, including less generative directions. The account also addresses incoming researchers and robotic materials. Building on this program-level perspective, the home+ case foregrounds the practical continuity of an assistive-robotics inquiry across physical rebuilds, researcher generations, and changing research conditions.

Gaver et al. characterize emergence as a feature of practice-based design research: methods, tactics, goals, and topics can change through engagement with materials and situations [6]. They advocate balancing responsiveness with intentionality rather than treating all unplanned change as productive. This perspective cautions against interpreting a long trajectory as the execution of an original roadmap. Our analysis examines which commitments remained recognizable and how particular developments enabled or constrained subsequent work, without treating the present robot as an inevitable endpoint.

2.3 Grounding Interpretations Across Designs

Gaver and Boucher's annotated portfolio connects ten designs from a two-decade program to interpretations of designing with data [5]. Its annotations remain connected to particular artifacts, providing a precedent for developing cross-project insights without detaching them from their evidence. Other retrospective accounts revisit earlier engineering research through subsequent experience [10] or articulate qualities across years of design experimentation [15].

The soma retrospective also explicitly grounds themes in multiple examples and excludes candidates for which evidence was insufficient [14]. Following this emphasis on grounding, our lessons must

be supported by documentary passages and attributed accounts, not merely by arranging prototypes in a timeline.

3 Approach

We examine the 2009–2026 trajectory of home+ through retrospective design-history analysis, combining documentary reconstruction with written accounts from long-term contributors. Informed by reflective design-program research [14], the analysis examines relationships among selected realizations rather than assuming a single sequence of progressively improved robots.

The authors occupy different positions in the history being reconstructed. The lead analyst joined the home+ team during the current high-fidelity phase and participates in developing that realization. Other researchers became involved more recently and therefore bring familiarity with the project without firsthand experience of its full trajectory. Long-term contributors provide firsthand knowledge of earlier stages and transitions. These roles provide complementary forms of access and distance: recent contributors can examine the earlier record without having participated in all of its decisions, while long-term contributors provide historical knowledge that is not fully recoverable from surviving documentation.

3.1 Sources and Reconstruction

The corpus includes publications, manuscripts, one master's thesis, technical project reports, a conceptual chapter, and prototype photographs and diagrams. Sources were assembled from the project's publication history and linked records. Selection was purposive, focusing on material that directly documents a realization, its design or evaluation, or its relationship to other work. Selection was iterative rather than fixed: as relationships among realizations became clearer during reconstruction, additional records were examined to confirm, qualify, or challenge those relationships. The resulting corpus was therefore developed through reconstruction rather than according to a predefined systematic-review protocol. External design-research literature provides interpretive context rather than evidence of events within home+.

For each selected instantiation, we record what was proposed, built, or evaluated; its research emphasis; reported constraints and limitations; and documented connections to other realizations. The source register preserves relevant versions and passages, distinguishing publication dates from construction or study dates. Unresolved build relationships remain qualified, and multiple reports of one study are not counted as independent evidence.

Historical footage in the supplementary video provides visual context, not a separately coded dataset. The current endpoint draws on the developer's account and the robot's URDF model, scaled meshes, and configuration files. The prototype is substantially implemented, with task-level tuning and user evaluation pending.

3.2 Written Retrospective Accounts

Two long-term project contributors, designated R1 and R2, provided written responses to twelve open-ended questions about personnel changes, handoffs, funding, institutional moves, unplanned opportunities, abandoned directions, and related builds. Both are coauthors and agreed to the use of their accounts and quotations.

An annotated draft timeline and early manuscript framed the request, and a demonstration video of the current prototype was supplied for context. The accounts therefore reflect both the contributors' historical involvement and the questions and materials presented to them. We compare their explanations with documentary evidence and with each other, retaining attribution and unresolved differences. Responses are identified by contributor and question number. The questions and written accounts are provided as supplementary material.

3.3 Analysis

The analysis developed iteratively alongside the reconstructed trajectory and manuscript. We compared selected instantiations in terms of embodiment, system scale, research emphasis, and fidelity, alongside contributor accounts of the research conditions surrounding those changes. A source-to-claim audit trail links candidate lessons to supporting passages, counterevidence, and unresolved questions, distinguishing documentary reports from retrospective explanations and our interpretations.

Candidate lessons were revised, narrowed, or withheld where the evidence did not support their proposed scope. Recurring patterns require support across relevant episodes, while a single account supports a bounded, attributed observation. The trajectory and build comparison summarize the reconstruction; the lessons articulate patterns across episodes, and the discussion considers their implications and limits.

AI assistance. ChatGPT supported question formulation, source retrieval and summarization, candidate-theme development and comparison, organization of the source-to-claim record, and manuscript drafting. Generated interpretations are working proposals, not primary evidence. The authors remain responsible for verifying sources and approving the final analytical claims.

4 The home+ Design Trajectory

Figure 2 summarizes selected home+ realizations from 2009–2026. The sequence includes both successive builds and overlapping research strands; publication dates do not necessarily indicate construction dates. Figure 3 provides selected mechanism and interface details.

Figure 2A: Initial concept [2009]. The comforTABLE concept was presented at HRI'09 as a networked assistive environment comprising a headboard ribbon, overhead display ribbon, and mobile side table that would adapt a person's immediate environment while supporting their activity [9].

Figure 2B: Early explorations [2009–2013]. Early physical explorations included a nightstand with a continuum table and learning-based control and a mobile nightstand integrating perceptual processes for navigation and interaction [1, 19].

Figure 2C: ART and clinical development [2014]. The Assistive Robotic Table (ART) concentrated development on a clinically oriented over-bed table and therapy surface. Its therapy surface supported wrist and hand exercises (Figure 3(a)), while clinical evaluation identified continuing maneuverability problems [8, 23].

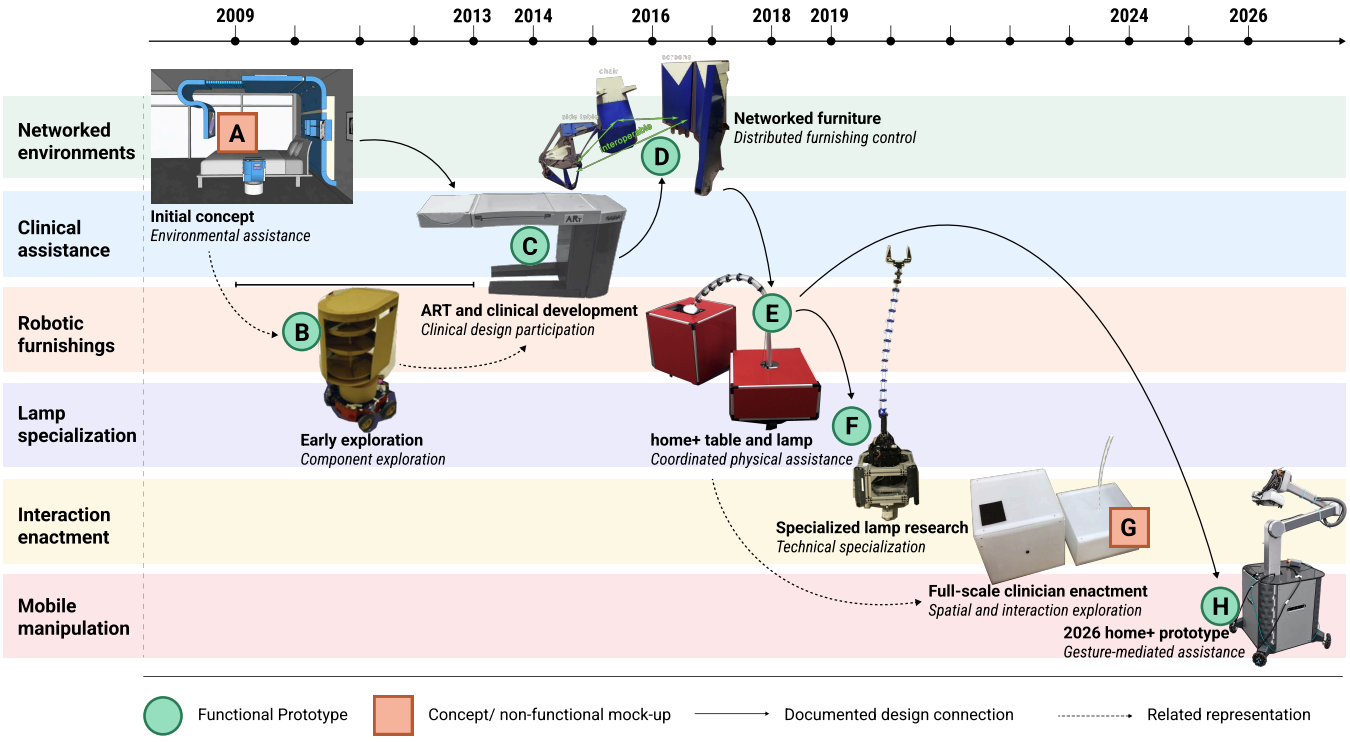


Figure 2: Selected realizations of *home+*, 2009–2026, arranged by research emphasis. Lettered markers locate dates in the record; B spans the stated period of early explorations. Solid arrows mark documented design connections, including continuation and redesign of the lamp strand; the dashed, undirected link compares functional furnishings with later full-scale interaction mock-ups. Circles identify functional prototypes; squares identify the concept and nonfunctional mock-ups.

Figure 2D: Networked furniture [2016]. The networked suite combined a lift-assist chair, morphing side table, and adaptive partition. The chair acted as the control hub, including gesture-based input for commanding surrounding furnishings (Figure 3(b)) [4].

Figure 2E: home+ table and lamp [2018]. The h+cube and h+lamp divided assistance between lower and upper workspaces. The cube used an internal lift, rotation, and vacuum gripper for floor-level retrieval (Figure 3(c)); laboratory demonstrations also explored cooperative object handling [24].

Figure 2F: A specialized lamp research strand [2019]. The continuum lamp, also known as CuRLE (Continuum Robotic Lamp Element), became a platform for configuration-space analysis and motion planning. The thesis documents successive lamp generations and redesign, while related work continued on other domestic home+ elements [11–13]. Table 1 summarizes selected technical changes.

Figure 2G: Full-scale enactment with clinicians [2024]. Full-scale table-and-lamp mock-ups without electronics were used with six clinicians to explore prospective interfaces, movement paths, and speeds through researcher-enacted behavior [2].

Figure 2H: The current home+ [2026]. The 2026 home+ prototype integrates camera-based gesture interaction with an omni-wheel mobile manipulator for household assistance, with particular attention to users with upper-limb mobility constraints. Coffee preparation is the current task context, with task tuning and user evaluation pending.

5 Lessons from the Long-Term Design Trajectory

The trajectory reveals six lessons about maintaining and revising a long-running design inquiry. Each connects particular episodes through documentary evidence and retrospective explanations.

5.1 A continuing inquiry can change its unit of design.

home+ remained connected as an assistive inquiry while its unit of design moved among environments, furnishings, and subsystems. This continuity is documented through explicit relationships among materially different realizations. The 2009 comforTABLE concept located assistance in a networked environment and emphasized supporting people's own activity [9]. The 2016 suite identified ART as a component of that wider ambition, while the 2018 robot-pair paper explicitly connected its design to ART, the patient-room concept, and the networked suite [4, 24]. These connections establish a relationship among the inquiries pursued through the artifacts, beyond their visual resemblance. The therapy

Table 1: Selected lamp implementations and the 2026 home+ model. L1–L3 follow the thesis's generation labels.

Characteristic	**L1: Earlier lamp** [11, 24]	**L2: Upgraded h+lamp** [11]	**L3: CuRLE** [11, 12]	**2026 home+** Model/configuration records and developer account
Base locomotion	Passive casters in the thesis's L1 account	Two encoder-equipped drive motors; differential drive	Differential drive with **wheel/caster suspension**	Four-wheel mobile base; omni-wheel arrangement reported by the developer
Arm structure	3 tendon drives; vertical lift	3 tendon drives; vertical lift	**4 tendon drives**; lift; **rotary base and wrist**	**4 revolute joints**: shoulder pitch, elbow pitch, wrist roll, hand pitch; no actuated lift modeled
Configuration / study scope	Direct control; no planning model specified in the compared account	Base planning: (x, y, θ)	Base (x, y, θ) and arm (ω, u, v); lift and wrist fixed in reported planning experiments	Base (x, y, θ) + 4 arm angles: **7 configuration variables**, plus gripper opening
End effector	Granular-jamming vacuum gripper	Motor-driven grasp; open-loop control	**Position-controlled grasp and wrist**; flexible finger attachments	Coupled two-finger gripper; **one independent opening coordinate**; modeled finger angles 0–90°
Control implementation	Direct wired switches in the thesis's L1 account	Arduino Mega; Wi-Fi shield; remote controller or queued paths	**Arduino Due; Raspberry Pi**; dedicated encoder-counter hardware	ROS 2/MoveIt model and planning configuration; custom serial execution bridge in the developer's records
Reported sensing	Feedback not specified in the compared description	Motor encoders	Encoders and **tendon-tension sensing**	Configured RGB-D camera roles: **D405** hand, **D435i** front, **D455** side
Reported use / status	Household-task demonstrations with researcher-operated controls	Teleoperation and preliminary base-planning tests	Configuration-space analysis and physical planning experiments	Gesture-mediated household assistance; task tuning and user evaluation pending

Notes. Tendon counts denote drives, not independent DOFs. Bold highlights selected characteristics.

surface and chair interface in Figure 3(a–b) illustrate two ways of pursuing this concern: supporting limb activity and enabling seated users to command surrounding furnishings [4, 8].

The object receiving concentrated design attention nevertheless changed. R2 associated ART's clinical emphasis with a lead researcher's interests and recalled a subsequent return toward the "elements of a robot-room" (R2, Q1). The clinical publication also documents requirements developed with clinicians, so personnel were not the only influence [23]. Describing movement between environmental and device-level concerns, R2 explained: "Both aspects were always present, but we tended to adjust to maximize the strengths and interests of the senior/lead students on the project" (R2, Q2). The account identifies a change in relative emphasis rather than the disappearance of one concern when another became prominent.

The continuum-lamp work likewise narrowed attention to a particular configuration-space and planning problem, while other home+ elements continued to be reported [12, 13]. Technical specialization was one strand within the broader program. Across these episodes, branching and re-convergence describe changes in the concentration of inquiry and renewed connections among concerns, not proven reintegration of every branch's hardware or software. An environment, furnishing, or subsystem could become the immediate unit of design without permanently defining the whole program. For HRI, the relevant connection is between the changing artifact and the human activity it is intended to support.

5.2 Re-embodiment combines inheritance with redesign.

Rebuilding a robot requires re-establishing inherited capabilities under the new physical conditions. Preserving a line of work did not require preserving its implementation unchanged. The 2018 paper reports that a manuscript served as a blueprint for a collaborating team to replicate the robot pair [24]. The subsequent thesis documents further physical changes as those systems became platforms for planning experiments. Table 1 summarizes selected build changes, while Figure 3(d–f) shows CuRLE's revised floor-contact, tendon-control, and grasping mechanisms.

The navigation experiments reveal why this redesign mattered. The upgraded h+lamp could receive and execute a planned sequence, but wheel velocity differences accumulated into positional error. On uneven surfaces, the passive casters could also prevent adequate driven-wheel contact. Reducing caster height reduced this problem but introduced rocking during acceleration, contributing to further drift. In the reported trials, the robot could collide with obstacles or leave the workspace rather than reliably reach its goal [11, Sec. 2.4]. Thus, executing planning commands was not equivalent to realizing the planned physical trajectory. For a furnishing intended to approach a person or retrieve an object, that difference affects whether the assistance can occur as intended.

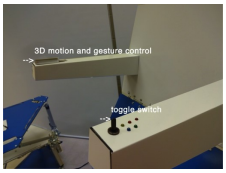
The subsequent rebuild introduced spring-loaded suspension for CuRLE's wheels and casters to address the contact and rocking problems (Figure 3(d)). Four tendon drives allowed bending control through opposing pairs, while tension sensing addressed slack that could make the arm's actual shape differ from the configuration inferred by its controller (panel e). Position-controlled grasping and an added wrist addressed uncontrolled closing and limited grasp orientation (panel f); a rotary base was also added [11, Sec. 3.3.1-3.3.4]. Yet a lift actuator was retained, and the mobile-base controller reused the same underlying software with retuned gains [11, Sec. 3.1, 6.1.1]. These changes addressed the conditions under which commands could be realized, rather than simply increasing the component count. The design response combined replacement with inheritance and required subsequent physical testing.

(a) Therapy surface



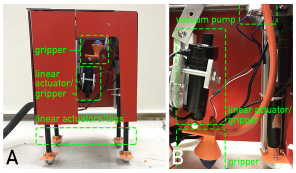
ART's surface supporting an arm. [8]

(b) Gesture-control interface



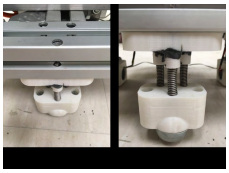
The chair's sensing and selection controls. [4]

(c) Internal pickup mechanism



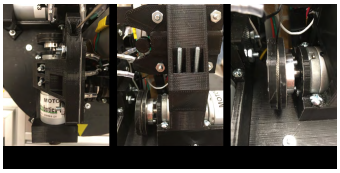
h+cube's lift, rotation, and vacuum gripper. [24]

(d) Revised floor contact



CuRLE's spring-loaded caster assembly. [11]

(e) Tendon-control hardware



Motor, spool, and tension-sensing assembly. [11]

(f) Grasp and wrist



Gripper views without and with flexible fingers. [11]

Figure 3: Mechanism and interface details complementing the whole-robot views in Figure 2. Panels (a–c) show distinct realizations of physical assistance and user control, not successive versions of one mechanism. Panels (d–f) show CuRLE details associated with the documented redesign in Table 1. Crops retain source annotations and photographic context. Panel (a) is a published therapy illustration, not a new evaluation.

Reconsideration also concerned the choice of mechanism. R1 emphasized retaining the project's ambition (R1, Q8), whereas R2 recalled moving away from highly compliant structures for elements requiring weight support or pushing (R2, Q8). These accounts concern different levels of commitment: an assistive ambition could persist while particular means of realizing it were revised. Abandoning or revising a mechanism could therefore serve the continuing assistive ambition. Re-embodiment involved deciding what needed to remain recognizable, what knowledge or components could be reused, and what required renewed design and testing.

5.3 Fidelity follows the question, not the technical stage.

Fidelity should be chosen for the question being asked, rather than inherited from the most technically complete prototype. ART's earlier process used cardboard representations and medium- and high-fidelity prototypes to investigate clinical requirements and interaction with the table [23]. The study reported in 2024 used full-scale table-and-lamp mock-ups without electronics after earlier work had demonstrated functional furnishings [2]. The paper identifies a specific limitation of earlier participant feedback: videos and three-dimensional models conveyed the concept but did not let people experience the furnishings' size, proximity, and presence in a domestic space [2]. The later study changed the representation to address that limitation, rather than simply repeating an earlier stage of technical development.

The mock-ups matched the intended furnishings in shape and size, while a researcher supplied their movement. Six clinicians considered hypothetical patient profiles and discussed appropriate interfaces, paths, and speeds. For some tasks, clinicians proposed how the furnishings should operate; the researcher enacted their proposal, and participants could revise it before another enactment [2]. The functional furnishings and the nonfunctional study representations are shown in Figure 2E and G, respectively. The clinicians' feedback included concerns about maintaining two controllers and whether automatically returning a mug to the kitchen would match the person's intention [2].

This procedure made prospective interaction available for inspection without requiring each proposed behavior to be implemented in software and hardware. The artifact's physical scale mattered to the inquiry, but autonomous functionality was not needed to enact the alternatives under discussion. We therefore interpret the change as a reduction in functional fidelity while retaining spatial information and enabling situated exploration. "Fidelity" did not move along one dimension.

The episode shows why a technically developed program may revisit questions through a less functional artifact. A functioning robot and a full-scale mock-up can make different aspects of the same assistive ambition investigable.

5.4 Handoffs combine reconstruction with redirection.

Preserving an assistive question does not automatically transfer the know-how needed to pursue it. R1 describes much of the work in their laboratory as passing through short-duration student generations, including many one-year master's projects and earlier brief periods of doctoral leadership (R1, Q2). Although outgoing students were expected to document their activities, results, and proposed next steps, R1 recalled that incomplete records required incoming researchers to act as "detectives" before advancing the work. For a physical robot, this includes the operating conditions behind demonstrated behavior, such as floor contact and controller tuning in the lamp experiments [11, Sec. 2.4, 6.1.1].

Student turnover also changed what could be pursued. R2 described adjustments to the strengths and interests of lead students (R2, Q2), while R1 associated a recent computer-science collaboration with expanded attention to machine learning and computer vision (R1, Q1). Earlier records already describe learning and perceptual processing [1, 19]; the later account therefore identifies a

change in collaboration and emphasis rather than the first appearance of those techniques. New participants could reinterpret the inquiry as well as reconstruct its technical state.

R1 offered that the absence of a sustained home+ doctoral-student lead in their lab made R1 more willing to continue the project with subsequent students (R1, Q5). R2's attribution of persistence to R1 points instead to continuing faculty stewardship alongside changing student responsibility (R2, Q3–Q4). These two roles address different needs: stewardship can propel a question worth pursuing, while handoff transfers the technical and contextual knowledge through which another researcher can pursue it. Informed reinterpretation requires both enough history to understand prior choices and enough freedom to reconsider them.

5.5 Opportunities enable continuation and redirect attention.

New opportunities shaped which part of a continuing assistive ambition could be pursued next. The two contributors describe a sequence of opportunities for continuation rather than the execution of a seventeen-year plan. R1 wrote, "I thought home+ had come to an end several times, but it keeps resurfacing" (R1, Q3), linking renewed activity to student interest and encouragement from a colleague. R2 emphasized the skills and interests of available students when recruitment became possible (R2, Q7). These accounts connect research direction to the people and resources through which the next period of work could take place.

Funding was described as enabling and redirecting that work. R2 associated pressure to propose new innovations in successive funding applications with a "choppy" trajectory (R2, Q5). R1, meanwhile, recalled maintaining a conceptual connection to home+ during the clinical branch, while R2 described fallow periods in which R2 expected the project to end (R1, Q4; R2, Q4). Their accounts distinguish conceptual commitment, active work, and expectations of continuation; these were not necessarily experienced in the same way by both contributors.

The history of home+ consequently resists a simple distinction between planned progress and arbitrary interruption. A recognizable inquiry allowed new opportunities to be taken up, but those opportunities influenced which concerns became actionable next. Continuation could preserve an ambition while changing the immediate agenda. The lesson is to examine what renewed activity enables and redirects, rather than equating survival with uninterrupted development of the same artifact.

5.6 Institutional moves require rebuilding to conduct HRI.

An inquiry can retain its vision while the local capacity to build robots and study interaction must be re-established. The publication record documents changing institutional affiliations and replication of the robot pair for concurrent experimentation at two laboratories [4, 24]. R1 described a move from adjoining laboratories to parallel work coordinated through regular remote meetings. Separate prototypes supported complementary efforts in motion planning and further design development (R1, Q6). R2 likewise recalled little change in the overall vision, but delays while laboratories and student teams were established after institutional moves (R2, Q6).

The need for another physical robot was not merely a matter of copying an artifact. R2 explained that systems were needed at both sites so engineering and human-factors work could proceed locally. Different budgets, expertise, and component availability shaped the builds, while experience with the earlier system motivated modifications (R2, Q12). R1 also described later concentration of work at one site as a collaborator's changing responsibilities reduced involvement (R1, Q6). Thus, the location of people and working robots affected which parts of the inquiry could be pursued together.

This differs from a student handoff or a hardware redesign: the challenge was to re-establish the combination of equipment, expertise, people, and coordination through which physical interaction could be investigated. For long-running HRI teams, an institutional transition therefore calls for deciding which capabilities must be available at each site, which can be shared remotely, and how complementary work will remain connected. Retaining the project vision alone did not provide that local capacity.

6 Discussion

The history of home+ suggests that continuity in long-term HRI is not simply about preserving artifacts or maintaining a project over time. We discuss what persists across these changes, how teams can support informed continuation, and why longevity does not establish assistive success.

6.1 Continuity has more than one object

Across the lessons, continuity operates at different levels: a recurring assistive ambition, particular technical capabilities, and the people and records through which knowledge is carried forward across researchers and sites. These do not necessarily persist together. An investigator can retain a commitment while incoming students reconstruct working knowledge. A new robot can retain control methods while replacing much of its hardware. A clinically focused branch in the research can remain connected to an environmental inquiry. The extended history makes these relationships visible across reports that address different immediate questions.

This reading complements programmatic accounts of changing commitments and experiments [14]. It also brings the question of reuse emphasized in *Interaction Composer* [7] into a different empirical context: in home+, inheritance concerns the conditions under which another research generation can make a related physical system workable. The distinction is not that home+ uniquely combines material and organizational change, but that its records expose their interaction across this particular history. A description of the robot alone cannot establish which knowledge remains available to those who subsequently take responsibility for it.

6.2 Designing for informed continuation

The case of home+ and its 17-year history suggests documenting project transitions as well as artifacts. The aim is to distinguish what can be inherited from what needs to be reconsidered when the embodiment, research question, or institutional setting changes.

The lamp's floor-contact problems motivate recording failure conditions, while reuse of its controller with different gains motivates distinguishing retained software from required retuning [11]. The researcher-controlled demonstrations and clinician enactments motivate recording who supplied control and which behaviors were represented rather than implemented [2, 24]. The institutional moves motivate recording access to working robots and local expertise, not just file locations (R1, Q6; R2, Q12). Together, these examples motivate preserving the conditions behind an interaction claim, not merely the files needed to operate a robot.

Records alone cannot ensure continuation. R2 proposed securing funds between awards so incoming students could overlap with departing researchers, noting that such overlap had sometimes occurred through circumstance rather than planning (R2, Q11). This recommendation highlights a distinction between preserving information and providing people with time and context needed to understand it. Strong student ownership of a contribution and continuing stewardship of a broader inquiry need not be alternatives.

Informed continuation also leaves research scope open to change. The clinician enactments show how an integrated, functioning robot need not be the appropriate vehicle for every subsequent question [2]. Conversely, feedback on an enactment cannot establish the feasibility of the behavior enacted. A transition between representations should make explicit which aspects of the earlier work are being carried forward, what the new representation makes possible to investigate, and what remains to be tested.

6.3 Longevity is not evidence of assistive success

A coherent retrospective can make the current home+ prototype appear to be the intended aim of all previous work. The contributor accounts instead describe opportunities, changes in emphasis, and uncertain continuation. Nor does a long sequence of prototypes establish that people received the intended long-term benefit. Cortellessa et al.'s assistive-robotics retrospective foregrounds technological, care-related, and human conditions for sustained use [3]. Those deployment concerns remain necessary questions for home+, not outcomes supplied by its longevity.

The distinction is between sustaining the capacity to investigate an assistive ambition and demonstrating that it has been met. The present history provides evidence of the former: what was retained, reconstructed, redirected, and made available for further work. Evaluating the latter still requires examining functional performance and the experiences of intended users in the contexts where assistance is needed. A long-term design account should make that remaining work visible rather than allowing the duration of a project to stand in for evidence of demonstrated benefit.

7 Limitations and Future Work

This study examines a single, selectively reconstructed design trajectory. The lessons are grounded in the history of home+ and do not claim to be representative of long-term robotics research more broadly. Without comparison cases, the analysis reported here cannot establish whether different staffing or funding arrangements would have produced greater continuity or different outcomes. The ownership and funding accounts describe contributors' experiences, not general advantages or funding-agency rules.

The available records are uneven in detail and do not preserve every decision, unsuccessful direction, or team-member handoff. Publication dates do not always establish when a prototype was built, and some relationships between builds remain unresolved. The parallel-build accounts cannot be matched to every photograph or used to overturn the published replication claim. Comparing sources supports particular claims but does not produce a complete chronology or causal account. The robot build comparison is descriptive rather than a standardized performance benchmark.

The retrospective accounts come from two long-term project contributors who are also coauthors. Their first-hand knowledge provides access to undocumented experiences, but their recollections may be shaped by hindsight and investment in the project. The annotated timeline supplied with the questions may also have guided which continuities and changes were emphasized. The accounts therefore do not independently validate our framing or represent the full range of contributor and stakeholder perspectives. No additional retrospective accounts were collected from former students or clinician collaborators. The 2024 study reports clinician judgments about enacted situations, not patient performance.

Finally, the seventeen-year span of home+ reflects a history of design activity rather than continuous deployment in homes. The current prototype still requires tuning and evaluation with users; the project's long history therefore should not be taken as evidence of sustained use or demonstrated benefit in the home. Future work could examine how continuity unfolds across ongoing and discontinued HRI programs while incorporating a wider range of team member accounts to identify which forms of knowledge, practice, and decision-making persist or change over time.

8 Conclusion

This paper traced the numerous milestones of home+, a research exemplar in human-robot interaction, across a seventeen-year trajectory. This tracing combined documentary records with written retrospective accounts to reconstruct how the project evolved over time. The analysis connects branching, technical specialization, and changes in design scale and prototype fidelity to a continuing inquiry into assistive robotic furnishings. It also shows how re-embodiment drew on knowledge retained with earlier iterations while requiring material redesign, and how team-member handoffs and new opportunities shaped the direction of the work. Institutional moves required re-establishing local capacity for building robots and studying interaction.

Together, the six lessons characterize continuity as something maintained across changing artifacts, people, and research conditions, rather than something guaranteed by preserving a single robot design or prototype. They foreground the costs of reconstruction alongside the possibilities opened by reinterpretation. For HRI, this perspective shifts attention from preserving a particular robot to preserving *the knowledge that enables its work to be reconstructed and reconsidered.* This includes not only functioning hardware and software, but also the rationale behind design choices, the questions that remain, and the knowledge needed to revisit those decisions as the embodiment or research context changes.

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